



Quarkus Skills

Agent Skills with Quarkus and LangChain4j

Community Central Theater Wednesday, May 13th 1:25 PM – 1:45 PM EDT

Ryan Jarvinen

Principal Developer Advocate

Red Hat / IBM

What do we all want?

Better tools and fewer problems!

Is AI the answer?

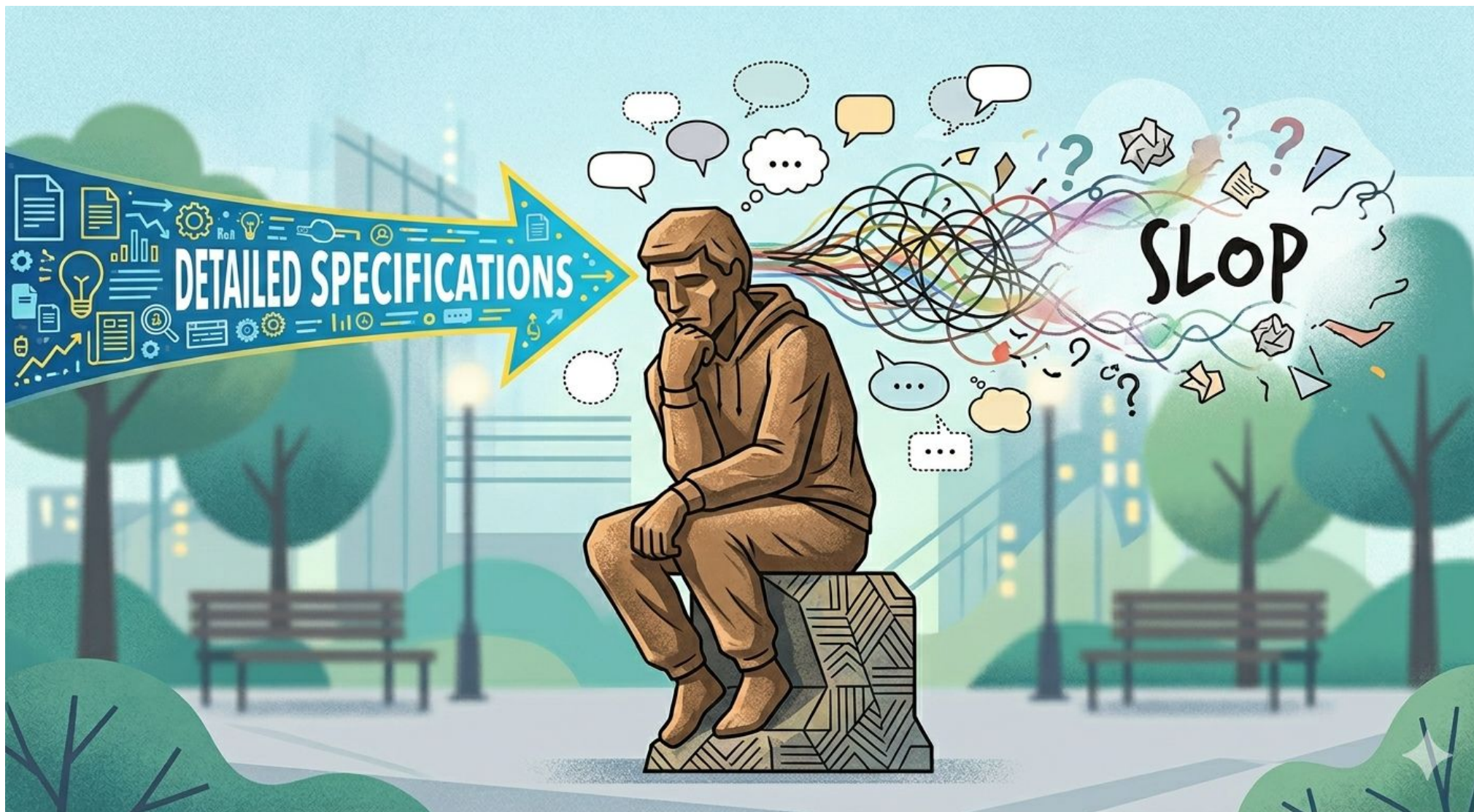
*I said **fewer** problems...*



What is the problem?

1. Context limitations and compression frequently lead to unreliable outcomes
2. Attempting to fix unreliable prompts by adding **more** detail or additional specs can result in **additional** context compression

Context Overflow



Agent Skills

<https://agentskills.io>

introduced by Anthropic in Dec 2025

Why skills?

- **Reusability:** Reuse workflows across multiple conversations
- **Consistency:** Apply the same approach for specific tasks
- **Specialization:** Create domain-specific expertise for reviews, testing, docs, and more
- **Flexibility:** Include supporting files like checklists, templates, and reference materials
- **Portability:** Skills use an open spec supported by many different Agentic services
- **Team collaboration:** Share standardized workflows with your team through version control

Skills spec: <https://agentskills.io/specification>

Each skill lives in a directory containing a **SKILL.md** file including:

1. Yaml Frontmatter:
 - **name** - lowercase letters, numbers, and hyphens (64 char limit)
 - **description** - what the skill does and when to use it (1024 char limit)
2. Markdown (prompt) - fewer than 500 lines

** Optional supporting files may be included in the skill directory*

Example:

```
---  
name: poem-writing  
description: Instructions for writing a poem  
---
```

When asked to write a poem, follow these rules strictly:

1. The poem **MUST** have exactly 12 lines.
2. The poem **MUST** be about a red hat.

Progressive Disclosure

Agents load skills through progressive disclosure in three stages:

1. **Discovery**: At startup agents load the entire skill into local memory, but sends only the **name** and **description** of each available skill to the LLM
2. **Activation**: When a task matches a skill **description**, the agent sends the full **SKILL.md** instructions and related files into the LLM context
3. **Execution**: The agent follows the instructions, optionally executing bundled code or loading referenced files as needed

Skills in Java with LangChain4j and Quarkus

LangChain4j and Quarkus Skills:

LangChain4j Skills docs:

<https://docs.langchain4j.dev/tutorials/skills/>

Quarkus Skills:

<https://docs.quarkiverse.io/quarkus-langchain4j/dev/skills.html>

Demo repo:

<https://github.com/ryanj/quarkus-skills-runner>

Using the `quarkus-langchain4j-skills` extension:

1. Add the `quarkus-langchain4j-skills` dependency to your `pom.xml`:

```
<dependency>
  <groupId>io.quarkiverse.langchain4j</groupId>
  <artifactId>quarkus-langchain4j-skills</artifactId>
</dependency>
```

2. In `application.properties` configure:

```
quarkus.langchain4j.skills.directories=.agents/skills
```

3. Annotate an `Assistant` to send a list of skills via the `SystemMessage`:

```
@RegisterAiService(systemMessageProviderSupplier =
SkillsSystemMessageProvider.class)
```

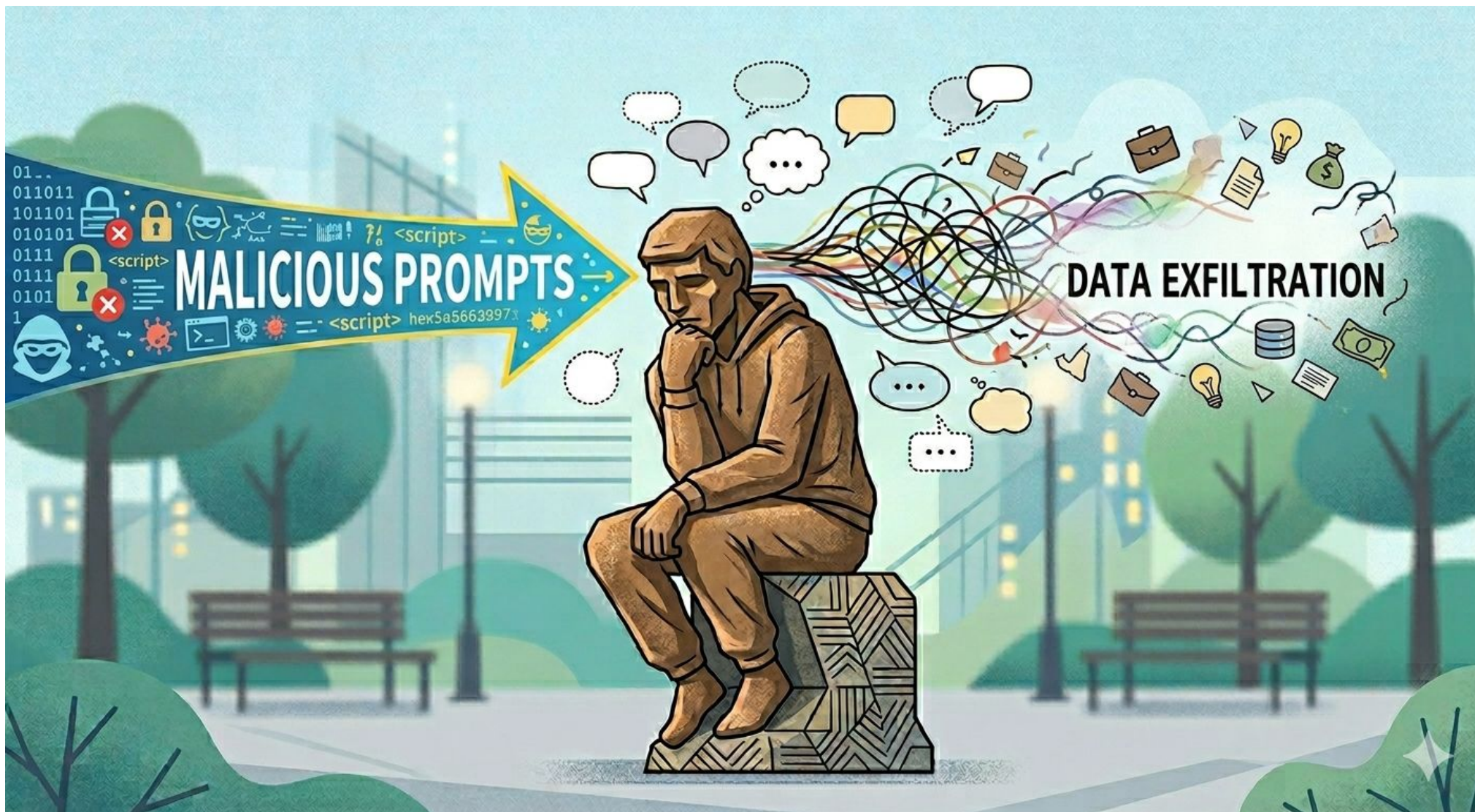
DEMO

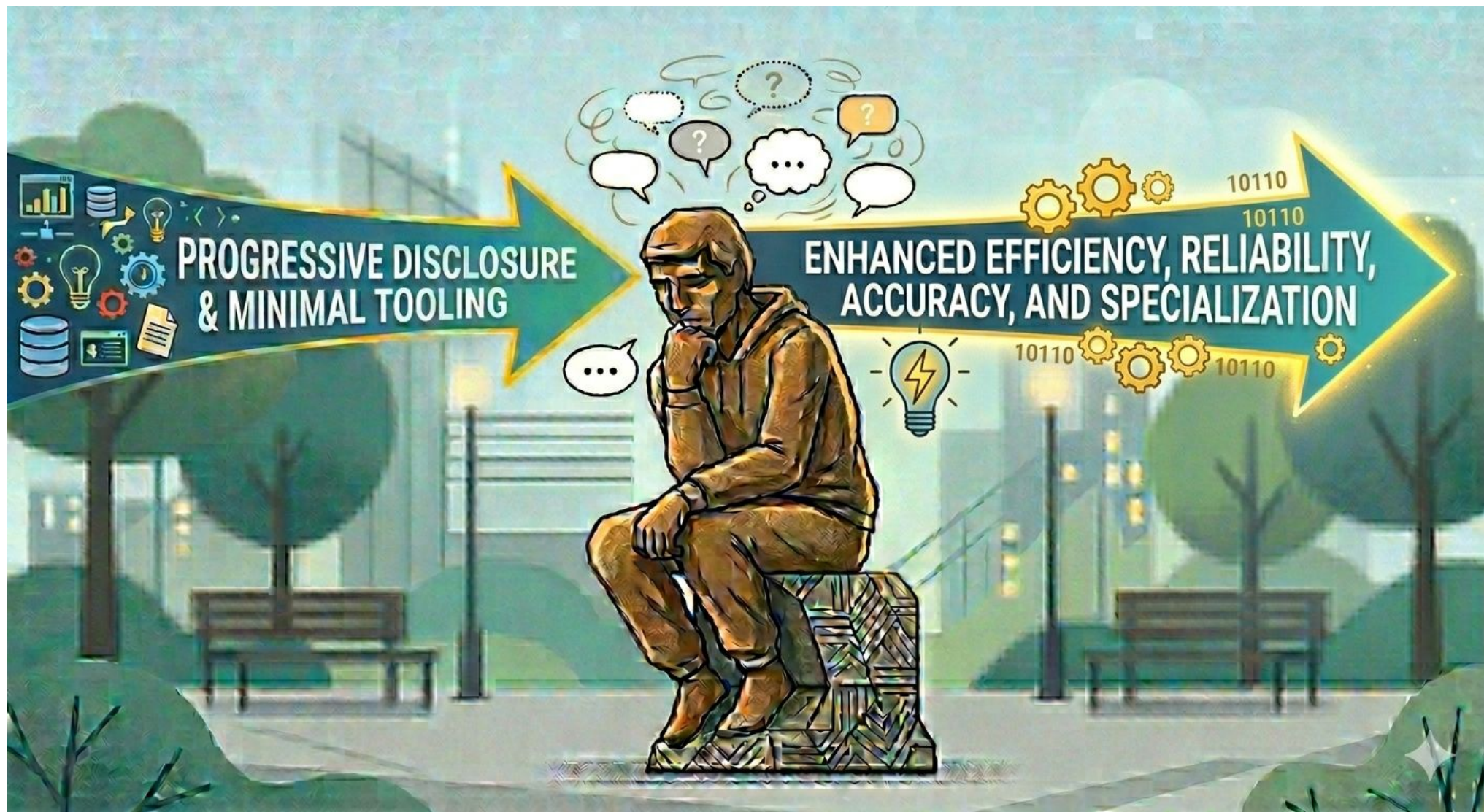
What skills? ⚠️

- Quarkus Skills: <https://github.com/quarkusio/quarkus-skills>
- Anthropic Skills: <https://github.com/anthropics/skills>
 - <https://skills.sh/anthropics/skills/skill-creator>
- Vercel Skills index: <https://www.skills.sh/>
 - <https://www.skills.sh/official>
- Karpathy Skills: <https://github.com/forrestchang/andrey-karpathy-skills>
- caveman Skills: <https://github.com/mattpocock/skills>

Usage Notes:

- Skills present a unique security risk. ***Use with caution!***
- Agent Skills depend on Tools to interact with the world
 - Skill portability may be limited by Tool availability
- Limit LLM tool access - Add *just enough* tooling for your requirements
- Quarkus allows Java functions to be easily exposed as Tools for capability, reliability, efficiency
- Use **Tool Guardrails** to protect Java function calls and to perform strict input and output validations





Are we there yet?

Next Steps

Links:

- Demo source: <https://github.com/ryanj/quarkus-skills-runner>
- Quarkus.io community website: <https://quarkus.io/>
- Quarkus Skills docs: <https://docs.quarkiverse.io/quarkus-langchain4j/dev/skills.html>
- LangChain4j Skills docs: <https://docs.langchain4j.dev/tutorials/skills/>
- Red Hat Build of Quarkus: <https://developers.redhat.com/products/quarkus>
- Markus Eisele's blog post on Skills with Quarkus:
<https://www.the-main-thread.com/p/java-agent-skills-kit>

Workshops:

- Red Hat Java 4 AI workshop (available on request)
- Quarkus LangChain4j workshop: <https://quarkus.io/quarkus-workshop-langchain4j>

Thank you!



Ryan Jarvinen

Principal Developer Advocate

Red Hat / IBM

rjarvine@redhat.com / rjarvine@ibm.com



Thank you

 linkedin.com/company/red-hat

 facebook.com/redhat

 instagram.com/red_hat

 x.com/RedHatSummit